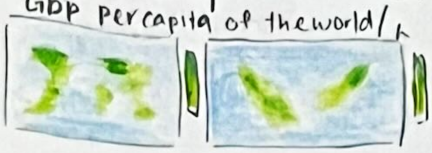
Title: Brainstorm
Author: Bernice Chua Chi Kheng
Student ID: 33589453
Sheet 1 out of 5

SHEET 1: BRAINSTORM

LAYOUT

① Choropleth map



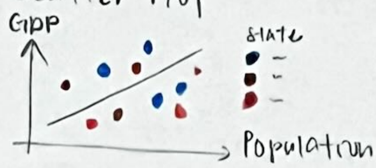
② Proportional Map

③ Pie Chart

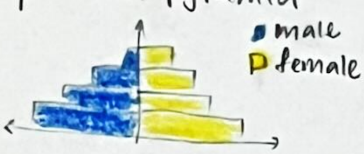
④ Bar Chart

⑤ Stacked Bar Chart

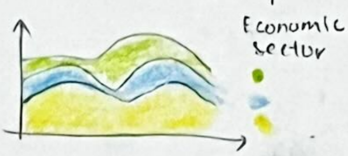
⑥ Scatter Plot



⑦ Population Pyramid

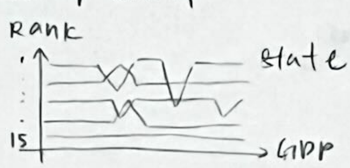


⑧ Stacked Area Chart

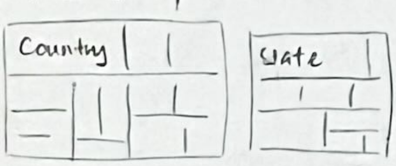


⑨ Doughnut Chart

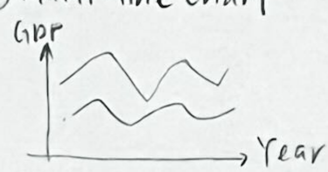
⑩ Bump Chart



⑪ Tree Map

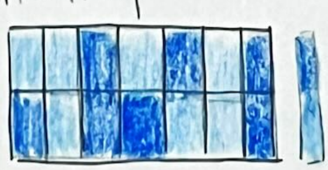


⑫ Multi-line Chart



⑬ Histogram

⑭ Heatmap



FILTER

Filter out ⑬, ⑭

⑬ irrelevant, not suitable in this case

⑭ irrelevant, not suitable in this case

CATEGORISE

①, ②, ④, ⑪			
5i, ⑦		5ii, ⑨	
⑥	⑩	③, ⑧	⑫

QUESTION?

1. Does this visualisation explore the GDP per capita of Malaysia, and any other related fields, from 2015 - 2024?
2. Will these charts intuitive or confusing?

COMBINE & REFINE

① & ② - Both show similar result, choropleth is better this case proportional symbol map more for absolute values, choropleth suitable for normalised values

SHEET 2: INITIAL DESIGNS

Title: GDP per capita of Malaysia (2015 - 2024)

Author: Bernice Chua Chi Kheng

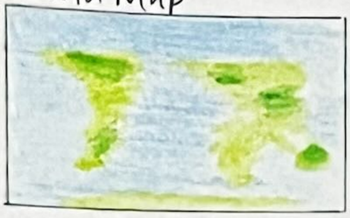
Student ID: 33569453

Sheet 2 out of 5

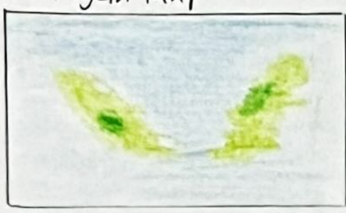
LAYOUT

① Choropleth Map

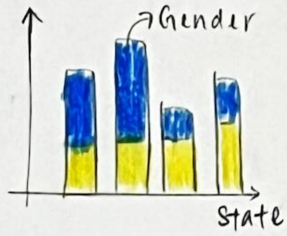
World Map

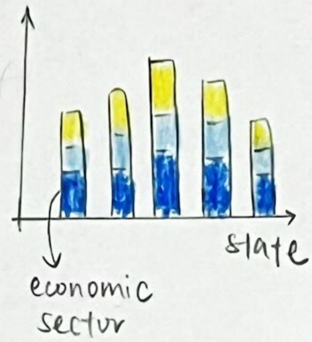


Malaysia Map



⑤ Stacked Bar Chart



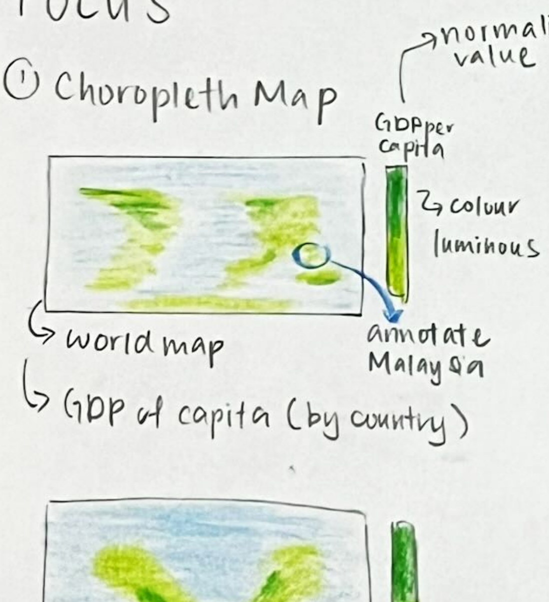


OPERATION

- Choropleth map
  - ↳ legend - colour luminous
  - ↳ annotation → label Malaysia
- stacked bar chart
  - ↳ legend - colour hue
- tooltip → howering over the map / chart will display the detail values (e.g., GDP per capita, country/state name)
- slider filter (2015 - 2024)
  - ↳ can see changes over years by dragging it

FOCUS

① Choropleth Map



↳ world map

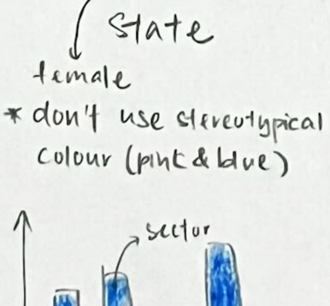
↳ GDP of capita (by country)

↳ Malaysia Map

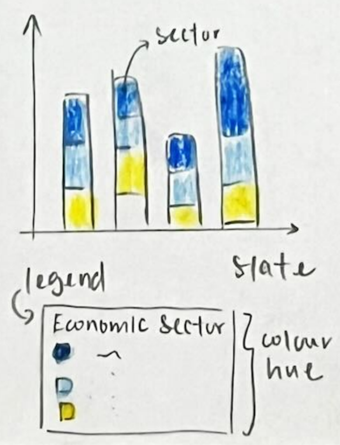
↳ GDP per capita (by state)

Year 2015 — 2024

⑤ Stacked Bar Chart



* don't use stereotypical colour (pink & blue)



DISCUSSION

Advantages:

- choropleth map
  - ↳ useful for visualising general patterns across geographical areas, and for showing how a measurement varies
- stacked bar chart
  - ↳ effectively show how a total amount is divided into parts, and can be space-saving

Disadvantages:

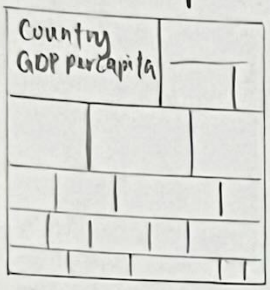
- choropleth map
  - ↳ assume the value within each region is uniform
- stacked bar chart
  - ↳ difficult to read for comparing individual segments that are not on the baseline

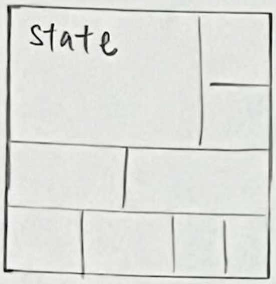
SHEET 3 : INITIAL DESIGN

Title: GDP per capita of Malaysia (2015-2024)
Author: Bernice Chua Chi Kheng
Student ID: 33589453
Sheet 2 out of 5

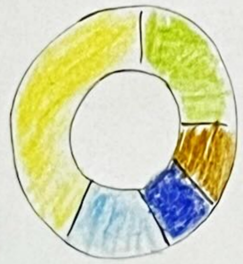
LAYOUT

(11) Treemap

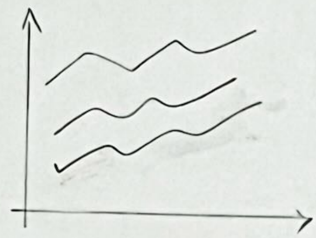




(9) Doughnut Chart



(12) Multi-line chart

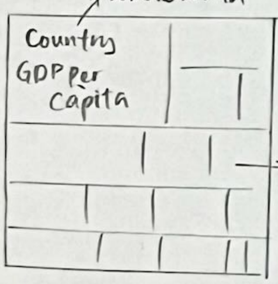


OPERATION

- tree map
- doughnut chart
  - ↳ colour hue for economic sectors – easier to identify
  - ↳ legend – user knows which colour represent which sector
  - ↳ annotation at the middle, reducing white space
- multi-line chart
  - ↳ annotate the states that are 2 highest GDP in 2015-2024.
  - ↳ annotate when the COVID-19 begins and check if it affects GDP
- tooltip
- filter — doughnut → state (→ slider filter); multi-line chart → state

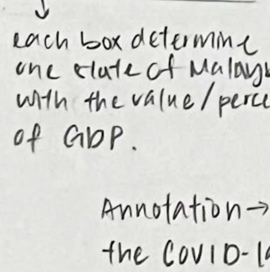
FOCUS

→ whole world



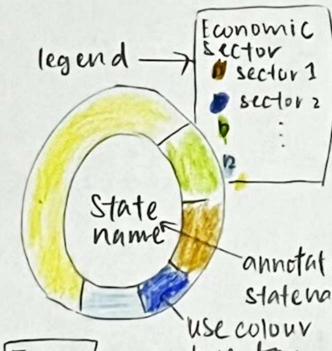
→ each box determine one country, with GDP value

→ malaysia



each box determine one state of Malaysia, with the value / percentage of GDP.

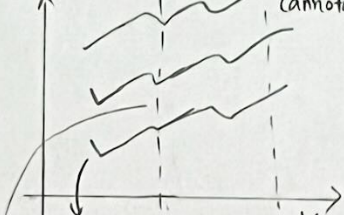
legend →



annotate statename

use colour hue for each sector

filter by states



each line represent each state in Malaysia

→ only use for 2 highest GDP states (others use grey → increase visual hierachy)

Annotation → label when the COVID-19 starts, and label which year has highest GDP

DISCUSSION :

Advantages:

- treemap → display data in a compact space, showing parts of a whole at a glance.
- doughnut chart → intuitive & simple for audiences to understand (few categories)
- multi-line chart → advantageous for comparing trends over time

Disadvantages:

- treemap → difficulty in comparing areas precisely
- doughnut chart → ineffective for too many categories
- multi-line chart → can become cluttered & difficult to read if too many lines are included

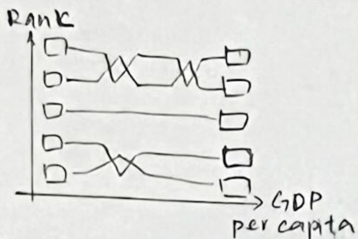
SHEET 4 : INITIAL DESIGN

Title : GDP Per Capital of Malaysia (2015 - 2024)
Author : Bernice Chua Chi Kheng
Student ID : 33589453

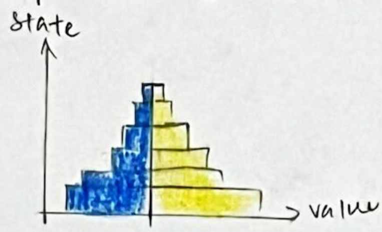
Sheet 4 out of 5

LAYOUT

(10) Bump chart

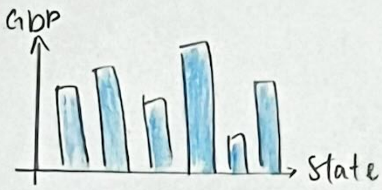


(7) Population Pyramid



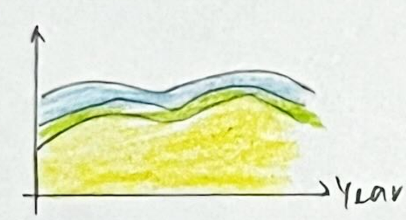
(3) Pie Chart

(4) Bar Chart



(6) Scatter Plot

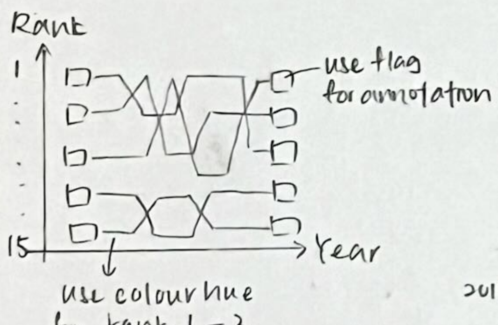
(8) Stacked Area Chart

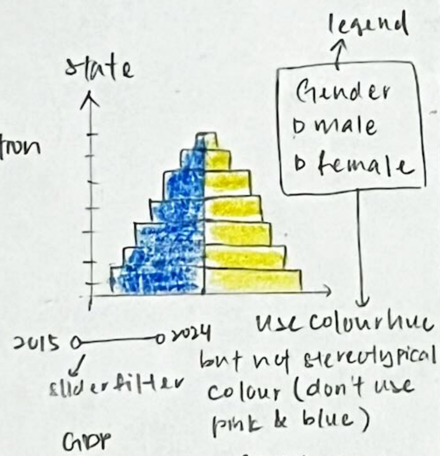


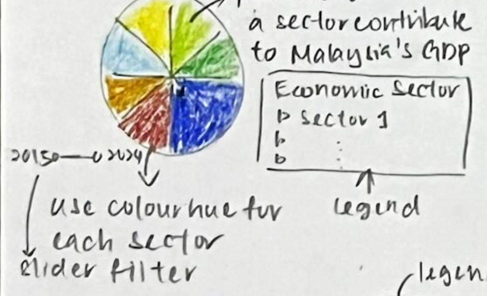
OPERATIONS

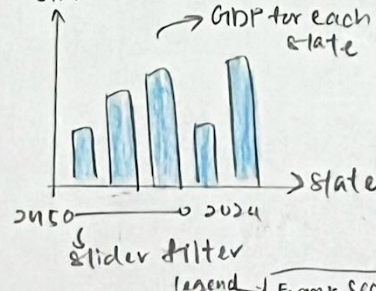
- bump chart → rank the GDP per capita for each state. highlight the TOP 3, increase visual hierarchy
- population pyramid
  ↳ filter by year (2015-2024)
- pie chart → percentage of each sector contribute to GDP
  ↳ filter by year (2015 - 2024)
- scatter plot → find relationship between GDP & Population
  ↳ filter by year & state
- stacked Area chart
  ↳ brushing interaction
- tooltip for each chart to view details

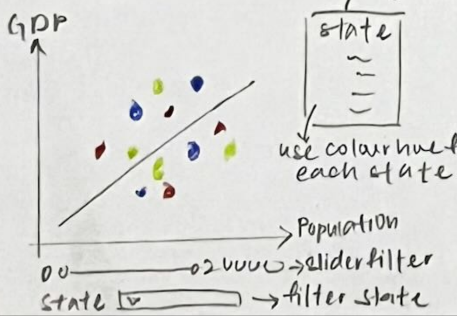
FOCUS

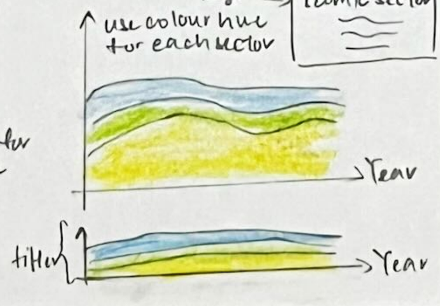












DISCUSSION

Advantages

- bump chart → clear display of how the ranking of different categories shifts over period
- population pyramid → show the population structure by gender
- pie chart → easy to understand
- bar chart → easy to compare data between categories
- scatter plot → visualising relationships between 2 variables
- stacked area chart → shows how different components combine to form a total, and changes over time

Disadvantages

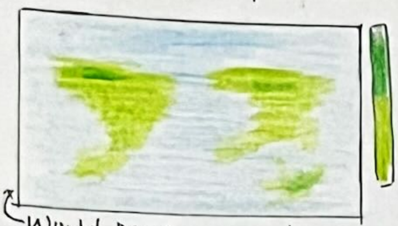
- bump chart → lack of details underlying values
- Population Pyramid & Bar chart → represents data at one specific moment
- Pie chart & Scatter plot → can be hard to intepret it too cluttered
- Stacked Area Chart → hard to read exact value

Title: Realisation
Author: Bernice Chua Chi Kheng
Student ID: 33589453
Sheet 5 out of 5

SHEET 5: REALISATION

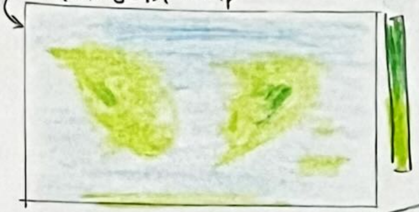
LAYOUT

1. Choropleth Map

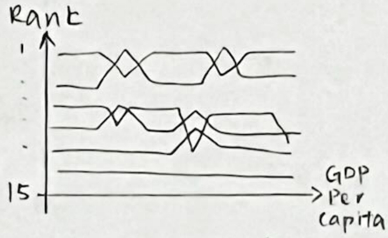


World Map

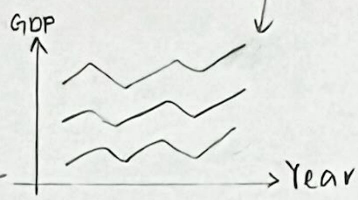
Malaysia Map



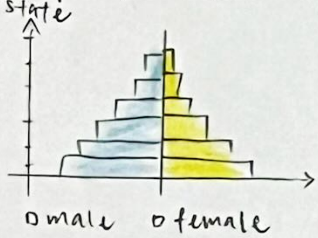
10. Bump Chart



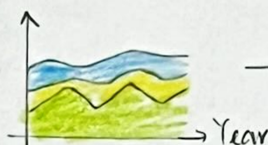
12. Multi-line chart



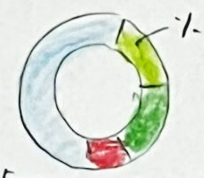
7. Population Pyramid



8. Stacked Area Chart



9. Doughnut Chart



OPERATIONS

- choropleth map
  - world
  - malaysia
- bump chart
- multi-line chart
- population pyramid
- doughnut chart
- stacked Area chart
- filters
  - slider
  - input (with choices)
- tooltips
- annotations

FOCUS

- explore the economic performance of Malaysia
  - ↳ GDP per capital } start from whole world, then focus on Malaysia
  - ↳ GDP
  - ↳ economic sector

world map

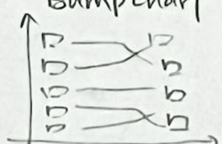
annotate where is Malaysia

GDP per capita of each country

malaysia map

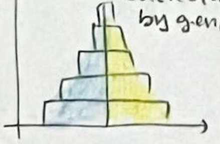
GDP per capita of Malaysia state

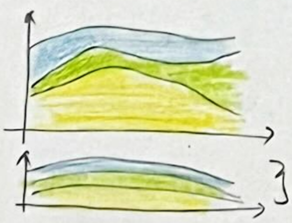
Bump chart



↳ Rank the states

explore the population for each state by gender





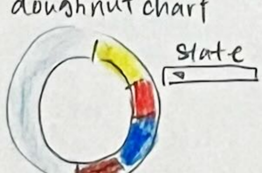
} brushing

explore each economic contribute to GDP

* use tooltip for all charts

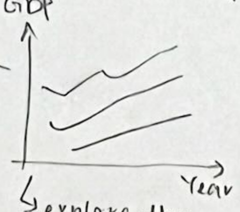
* ensure all the charts can explore from 2015 - 2024

doughnut chart



select the state, and find out the percentage of each sector contribute to GDP

Multi-line chart



↳ explore the GDP for each state

DETAILS

- datasets
  - ↳ mainly from World Bank, satistic.gov.my & DOSM
- dependencies
  - vegalite
    - ↳ json
    - ↳ html
  - python / R → data cleaning & combining
- estimate
  - ↳ data cleaning: 1 - 2 days
  - ↳ visualisation: 1 week
  - ↳ interactivity & filters: 1 day
  - ↳ refinement: 1 day